EE 354/CE 361/MATH 310 -

Introduction to Probability

and Statistics

Fall 2025

EE 354/CE 361/MATH 310 L2 section (Fall 2025)

Introduction to Probability and Statistics -

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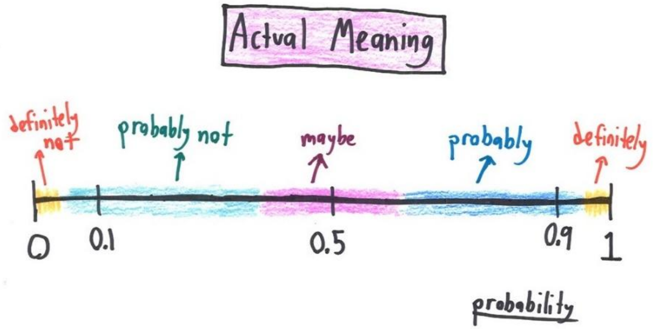
Week 11 - Lecture No 20 – October 27, 2025



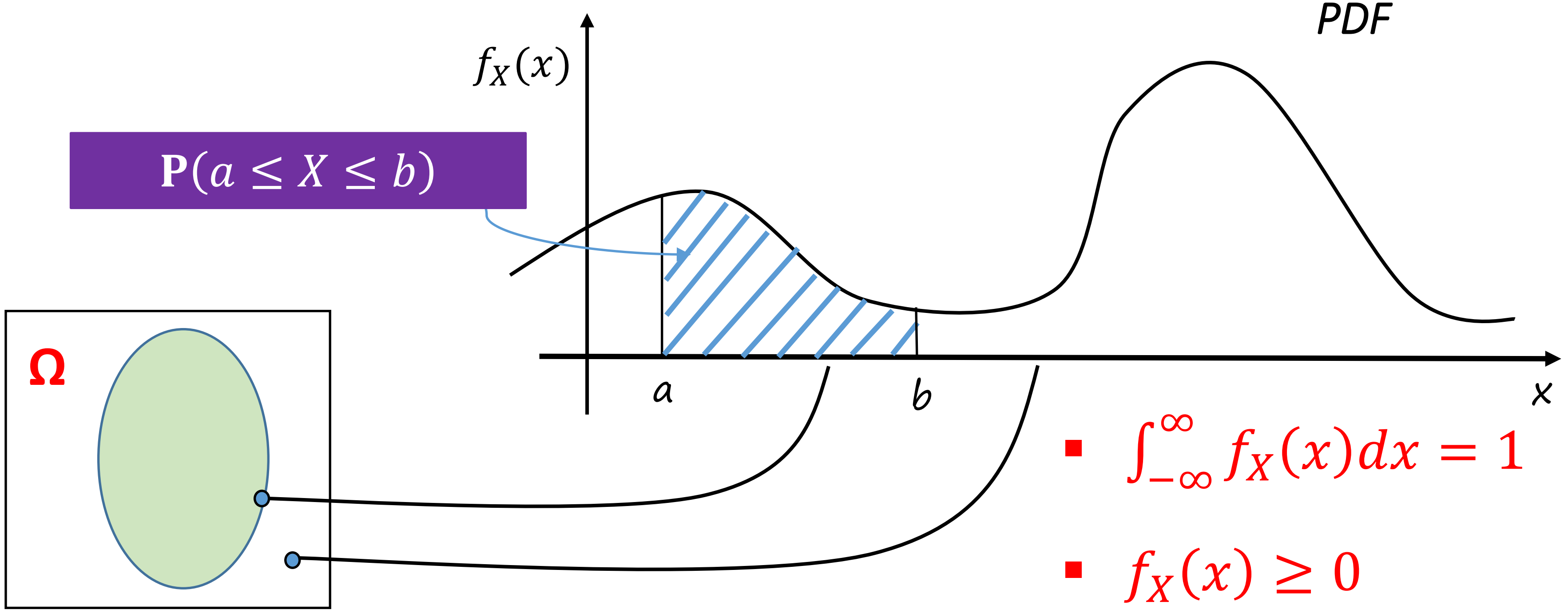
Announcements!

1. All lectures and grades (except Quiz No 7) posted.
2. Assignment No 02 uploaded – ***due date November 10, 2025*** (*Assignment 3 to be uploaded by 2nd week of November*)
3. Midterm copies – (last day for discussion & correction (if any) – ***October 28, 2025***)

Agenda for today

1. Unit 5 – Continuous Random Variables
    a. Recap – Probability Density Function
    b. Cumulative Distribution Function (CDF)
    c. Examples of few continuous Random Variables
2. Quiz No 06 (*towards the end – the lecture might help you solve the Quiz*)

Probability Density Functions (PDFs) – Recap - 1



- $\int_{-\infty}^{\infty} f_X(x)dx = 1$
- $f_X(x) \geq 0$

Definition: A random variable is continuous if it can be described by a PDF

$$\mathbf{P}(a \leq X \leq b) = \int_a^b f_X(x)dx$$

$$\mathbf{P}(X = a) = 0$$

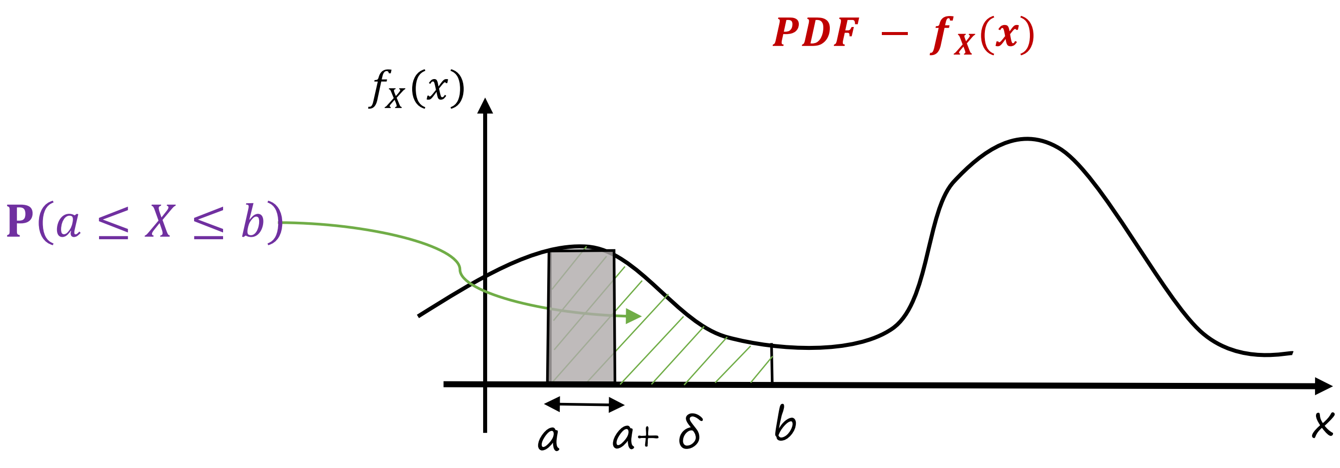
Aside - $P(1 \leq X \leq 3 \text{ or } 4 \leq X \leq 5) = P(1 \leq X \leq 3) + P(4 \leq X \leq 5)$

Probability Density Functions (PDFs): Recap - 2

$f_X(a)$??

$\delta > 0$, small (approaching zero)

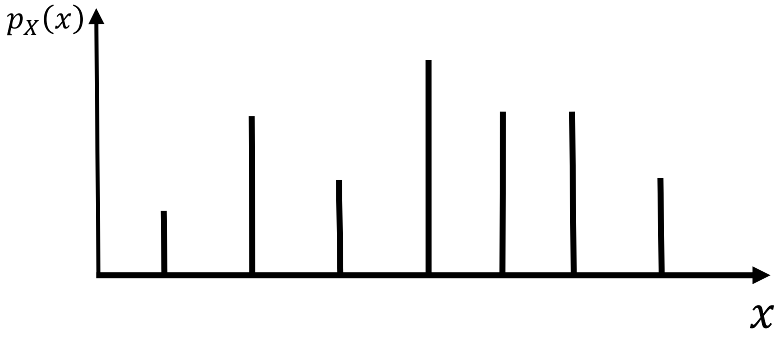
$$\boxed{\mathbf{P}(\mathrm{a} \le \mathrm{X} \le a + \delta) \approx f_X(a)\delta}$$



- $\int_{-\infty}^{\infty} f_X(x)dx = 1$
- $f_X(x) \ge 0$

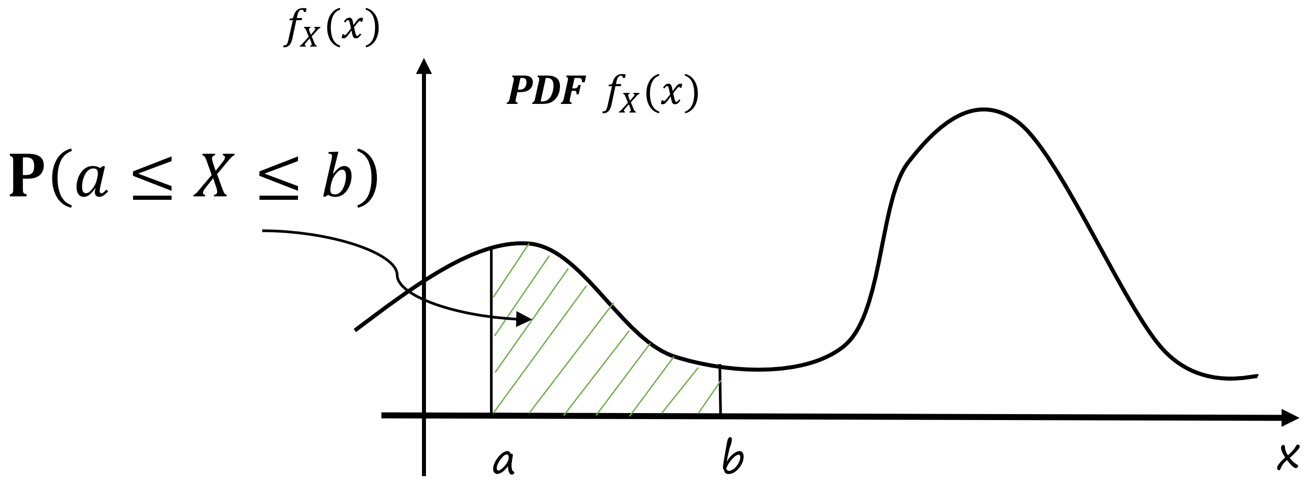
$$\mathbf{P}(\mathrm{a} \le \mathrm{X} \le \mathrm{b}) = \int_a^b f_X(x)dx$$

Expectation of a continuous RV



$$E[X] = \sum_x x\, p_X(x)$$

- **Interpretation**: Average in large number of independent repetitions of the experiment



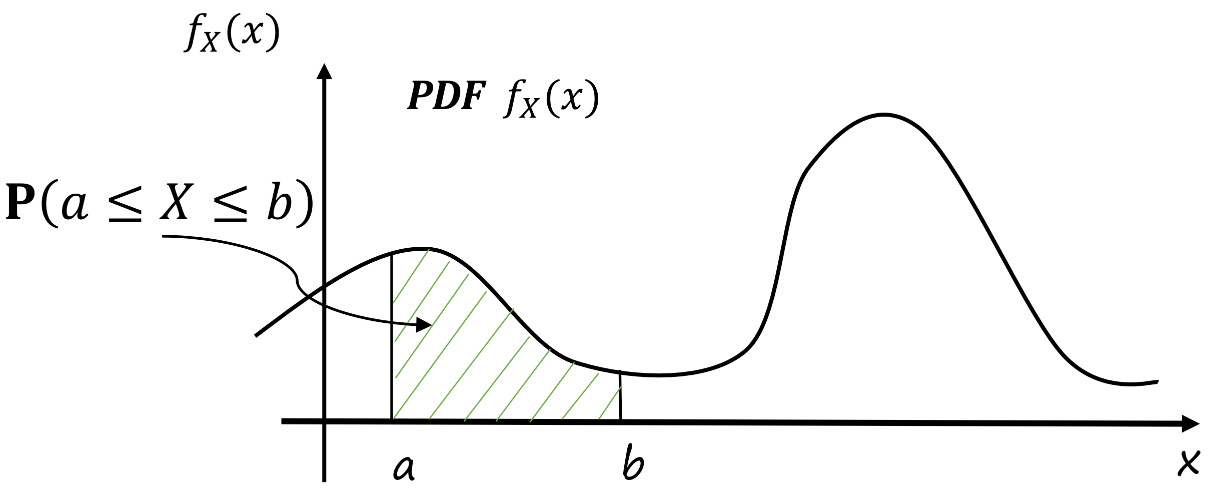
$$E[X] = \int_{-\infty}^{\infty} x\, f_X(x)\, dx$$

Assume, $\int_{-\infty}^{\infty} |x| f_X(x)\, dx < \infty$

Expectation of a continuous RV

Assume, $\int_{-\infty}^{\infty} |x| f_X(x)\, dx < \infty$?

A continuous RV with an infinite mean is one for which the expected value, calculated by an integral (A), diverges. *The most common example is the Pareto distribution, where PDF has a "long tail" that extends indefinitely, causing the weighted average to become infinitely large*



$\Rightarrow E[X] = \int_{-\infty}^{\infty} x\, f_X(x)\, dx$ A

Example: Pareto distribution

- The Pareto distribution is a classic example used in fields like economics to model phenomena like wealth or income.
- Its probability density function (PDF) is $f(x) = \frac{\alpha}{x^{\alpha+1}}$ for $x \geq 1$ and $\alpha > 0$.
- The mean of this distribution is the expected value, calculated by the integral $\int_1^{\infty} x \cdot f(x) dx$. A
- This integral only converges to a finite value if $\alpha > 1$. When $\alpha \leq 1$, the integral diverges to infinity. This means the average value of a Pareto-distributed variable is infinite if the parameter α is 1 or less.

Properties of Expectations

- If $X \geq 0$, then $\mathbf{E}[X] \geq 0$
- If $a \leq X \leq b$, then $a \leq \mathbf{E}[X] \leq b$
- Expected value rule:

$$\boldsymbol{E}[g(X)] = \sum_x g(x) p_X(x)$$

$$\boldsymbol{E}[g(X)] = \int_{-\infty}^{\infty} g(X) f_X(x)\ dx$$

$$\Rightarrow \quad E[X^2] = \int_{-\infty}^{\infty} x^2 f_X(x)\, dx$$

- Linearity

$$\boldsymbol{E}[aX + b] = a\boldsymbol{E}[X] + b$$

Variance and its properties

Definition of Variance: var(X) $= \boldsymbol{E}[(X-\mu)^2]$

$\mu = E[X]$

- Calculation using the expected value rule: $\boldsymbol{E}[g(X)] = \int_{-\infty}^{\infty} g(X) f_X(x)\ dx$

$$\text{var}(X) = \int_{-\infty}^{\infty} (X-\mu)^2 f_X(x)\ dx \qquad g(X) = (X-\mu)^2$$

$\text{var}(aX + b) = a^2\, \text{var}(X)$

Standard Deviation: $\sigma_{(X)} = \sqrt{\text{var}(X)}$

A useful formula: $E[X^2] - (E[X])^2$

Cumulative distribution function (CDF)

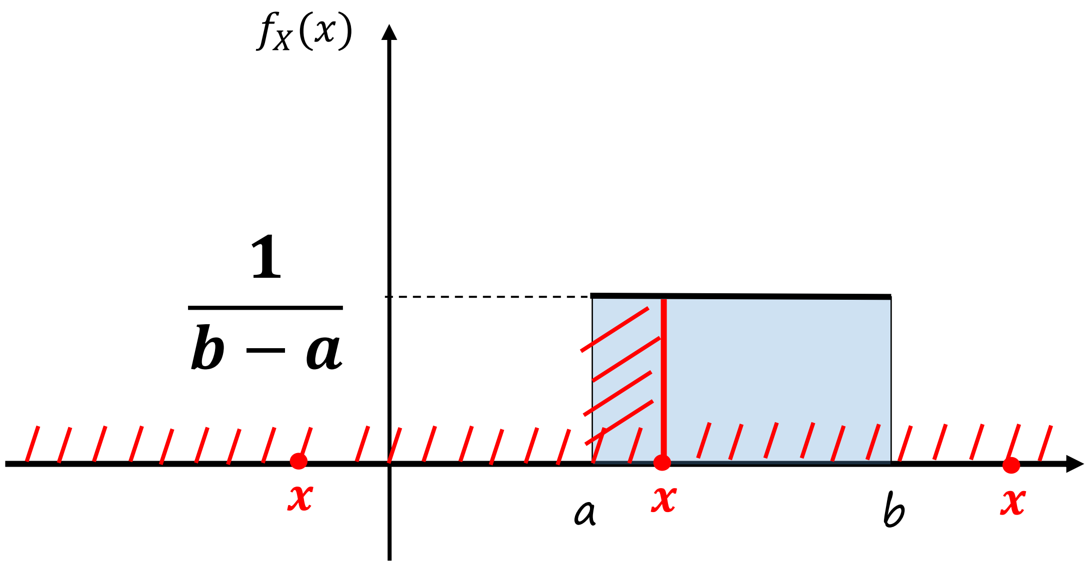
CDF definition: $\boldsymbol{F_X(x) = P(X \leq x)}$

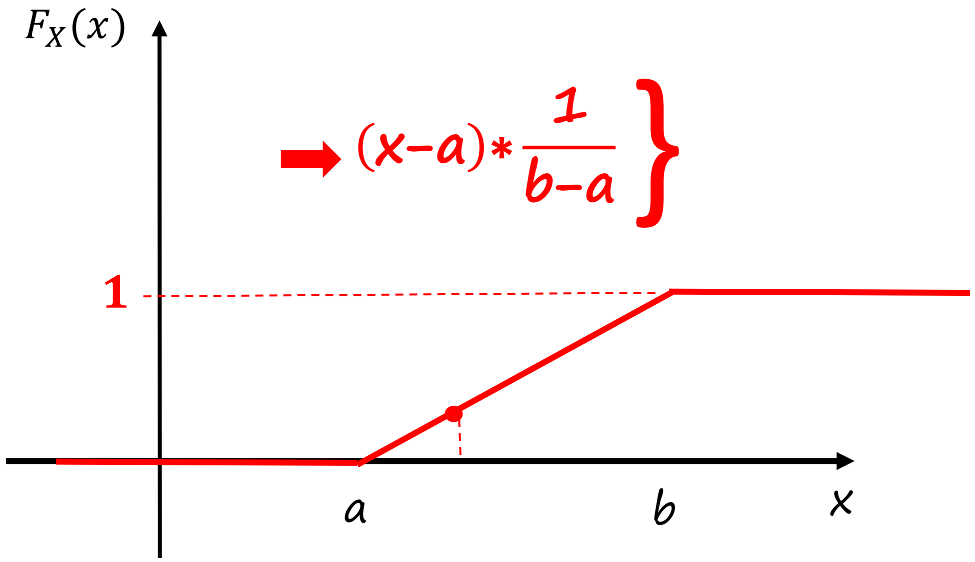
- Continuous random variables:

$$F_X(x) = p(X \leq x) = \int_{-\infty}^{x} f_X(t)\,dt$$

$\Rightarrow P(X \leq 4) = P(X \leq 3) + P(3 < X \leq 4)$

$\Rightarrow \boxed{\dfrac{dF_X}{dx}(x) = f_X(x)}$





Cumulative distribution function (CDF)

CDF definition: $F_X(x) = P(X \leq x)$

- Discrete random variables:

$$F_X(x) = p(X \leq x) = \sum_{k \leq x} p_X(k)$$

$p(X \leq 1) = 1/4$

$p(X \leq 3) = 3/4$

$p(X \leq 1) = 1$